

Lecture 15 – Bioenergy and Geothermal

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Outline

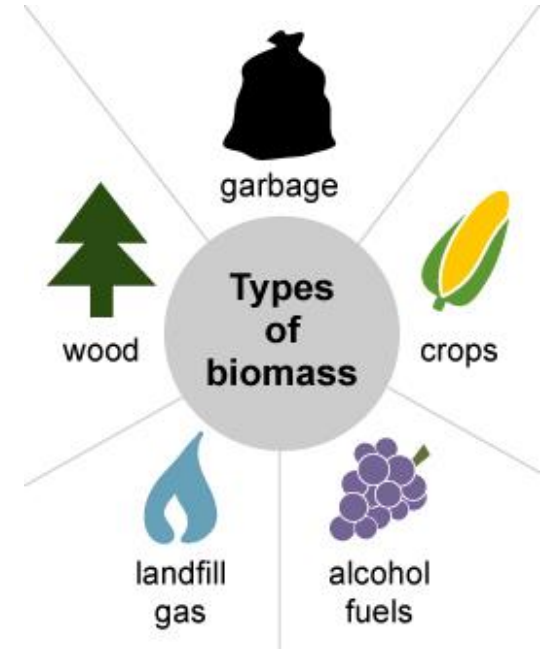
- Biomass Sources and Plants
 - Biomass Conversion
 - Biomass Statistics
 - Geothermal Energy Source
 - Geothermal Energy Technologies
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- B3: Chap 5 and 10
 - B1: Sec 8.7-8.8
 - REN21: Renewables 2024 Global Status Report

Bioenergy



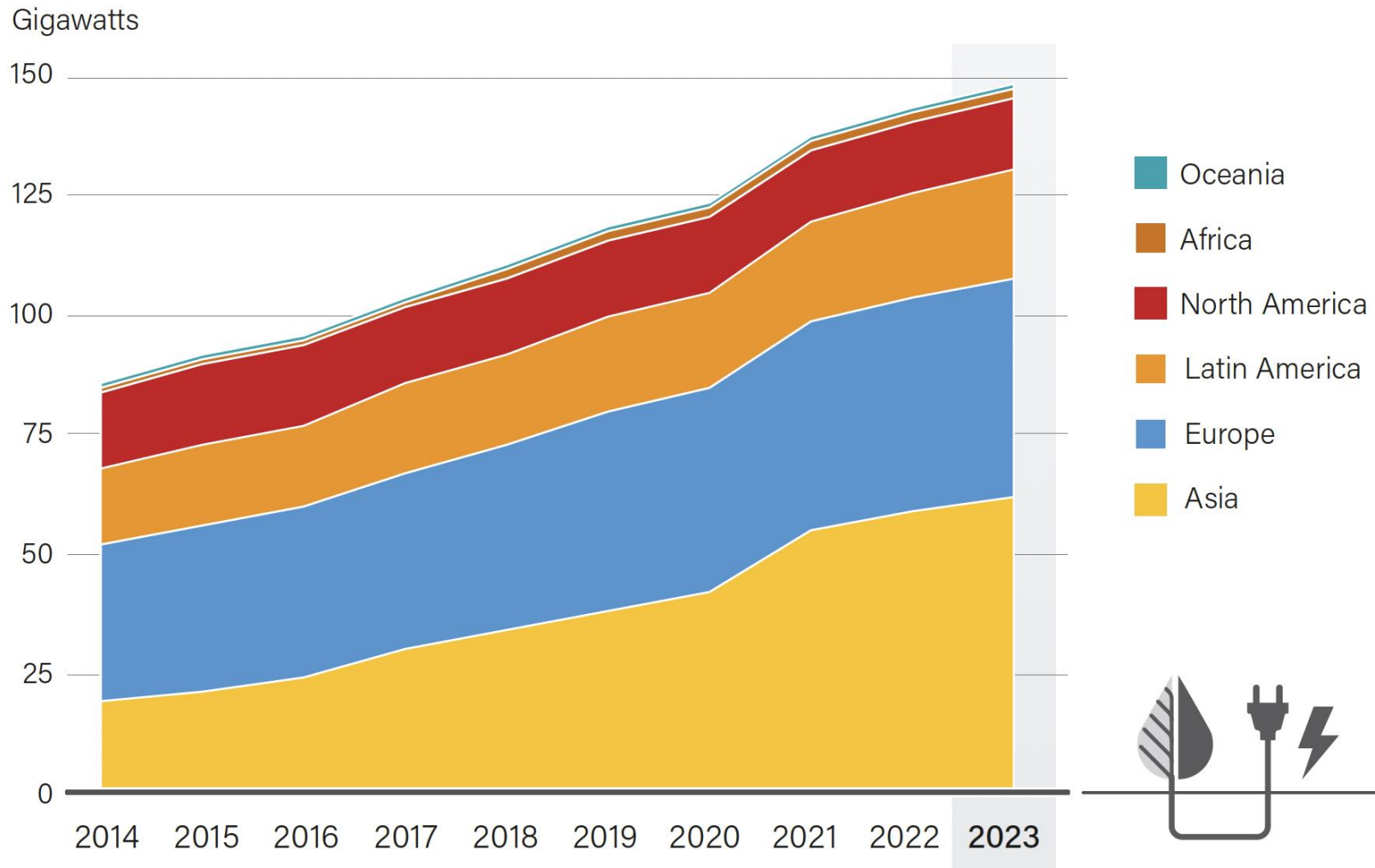
Biomass

Biomass: Getting energy by burning wood and other organic matter



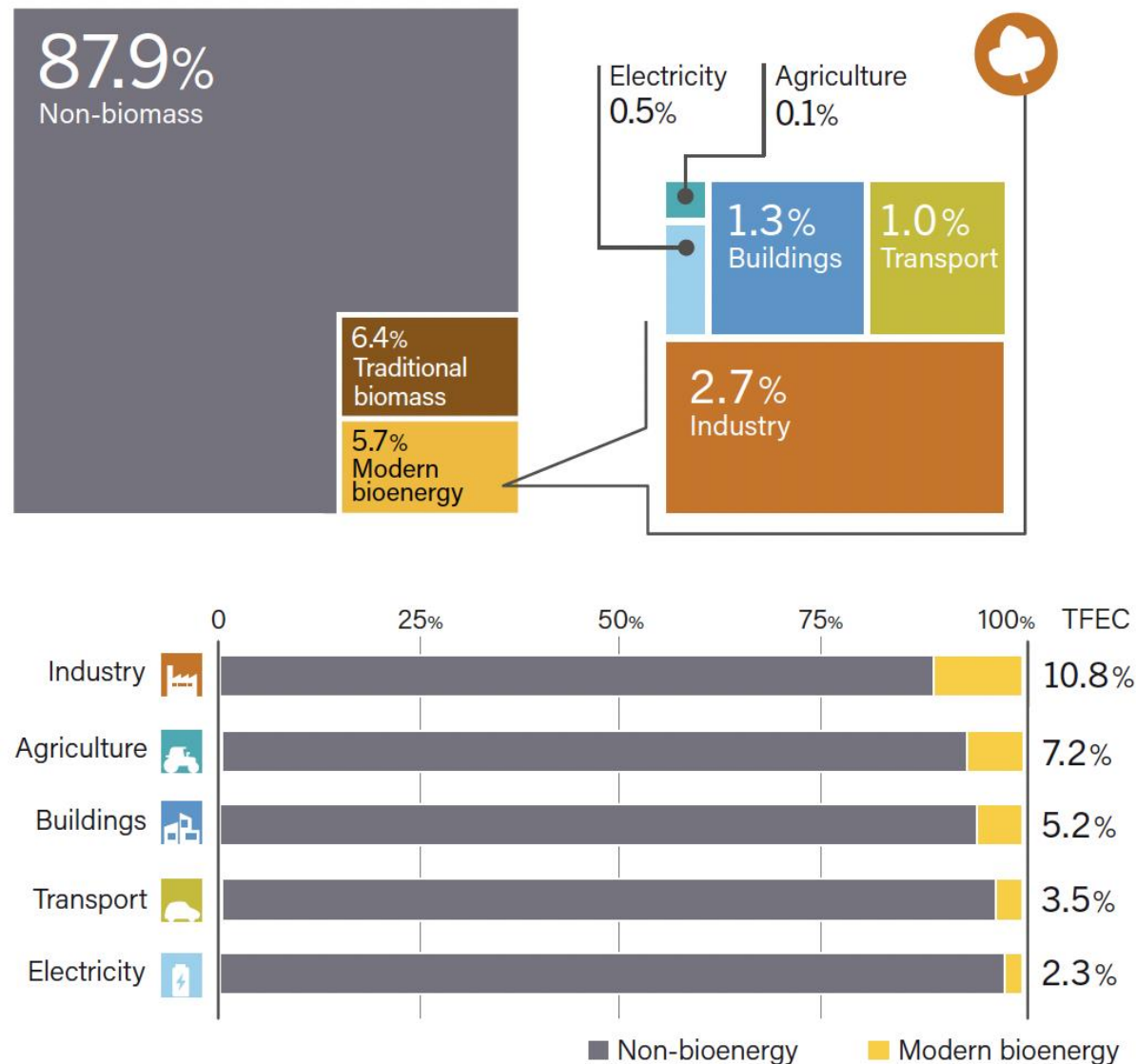
Biomass – Global Development (1/2)

Global Bio-Electricity Installed Capacity, by Region, 2014-2023



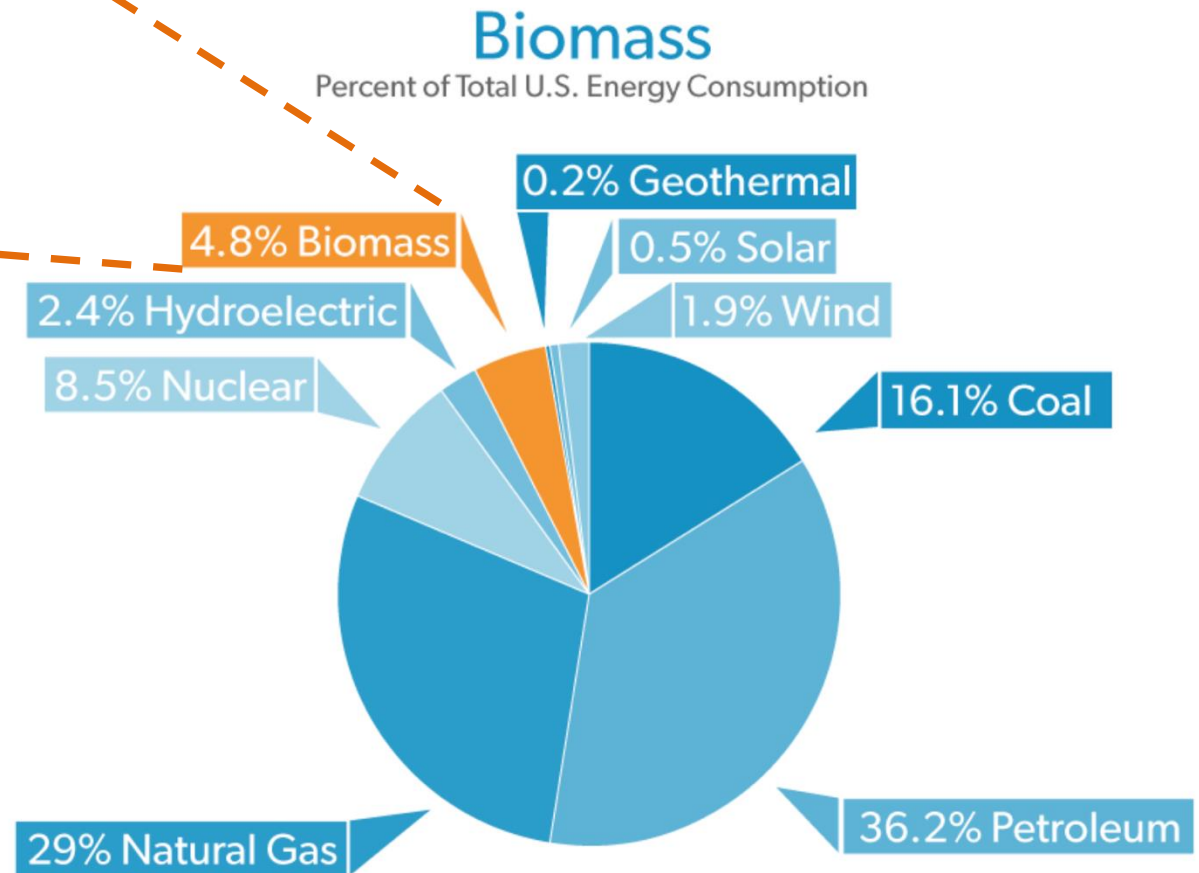
Biomass – Global Development (2/2)

Share of Bioenergy in Total Final Energy Consumption (TFEC), by End-Use Sector, 2021



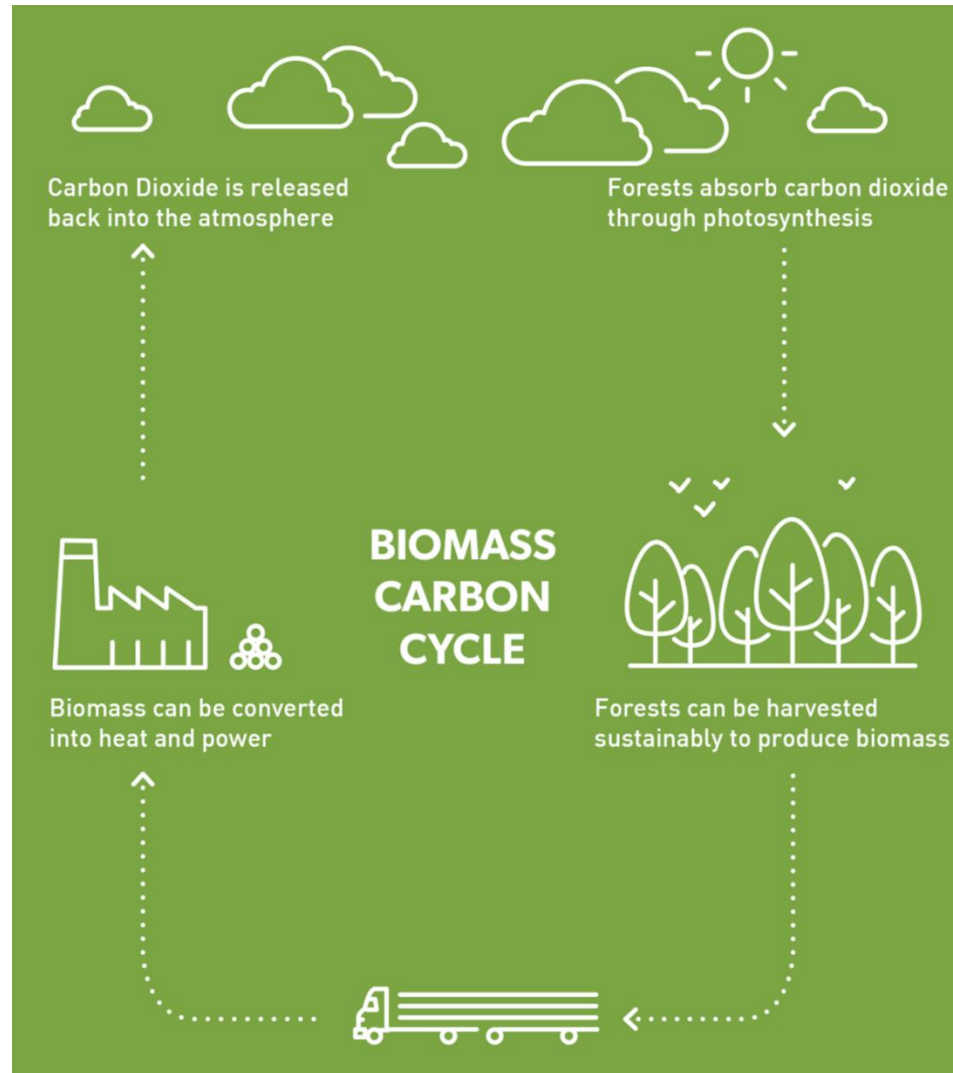
Biomass – U.S. Status

- 48% – biofuels (mostly ethanol)
- 41% – wood based biomass
- 11% – municipal waste

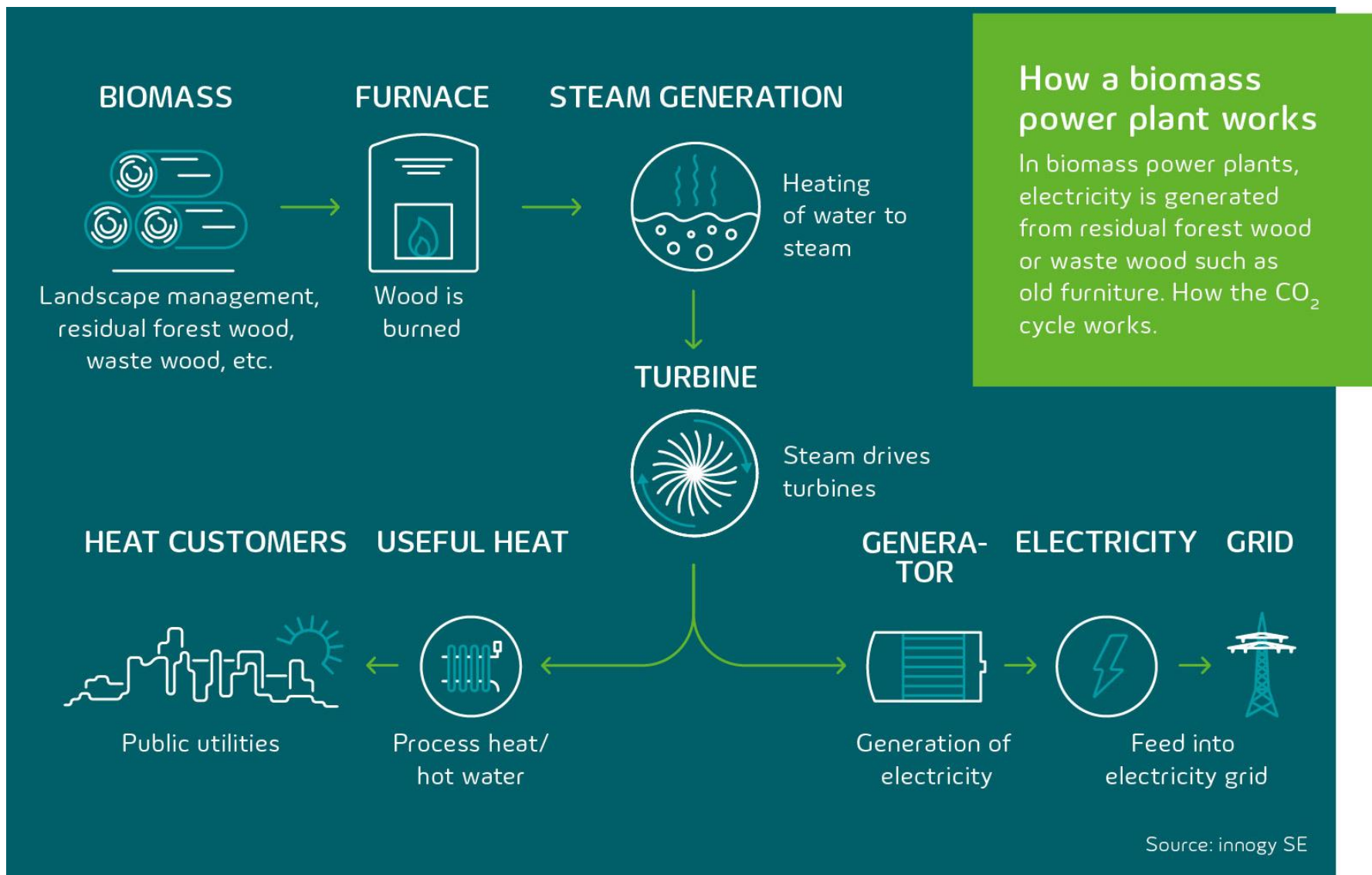


Biomass Carbon Cycle

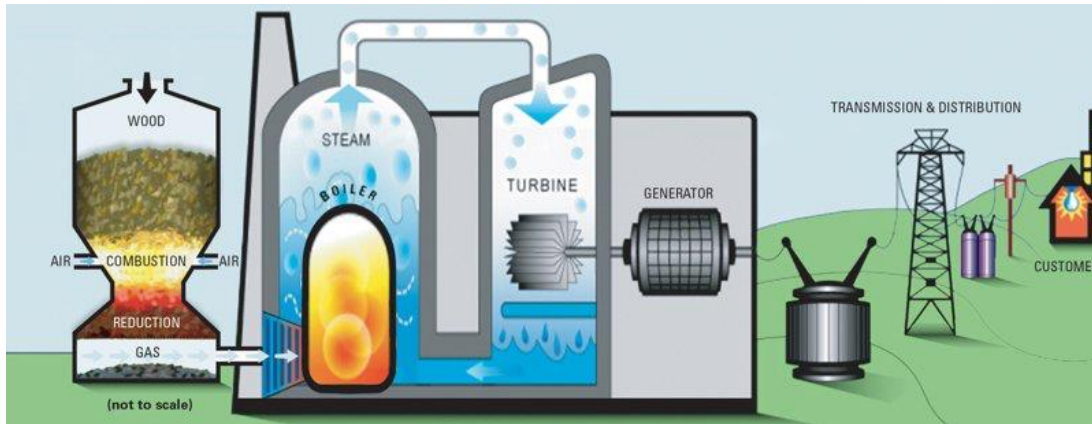
The biomass carbon cycle shows the roles of sunlight and photosynthesis and the resultant power generated by a biomass power plant



Biomass Power Plant



Biomass Power Plant (cont'd)



- 1) Moisture is first driven off.
- 2) At 400°C, the biomass begins to break down
- 3) Yielding a product gas (syngas) & solid byproducts (char and ash).

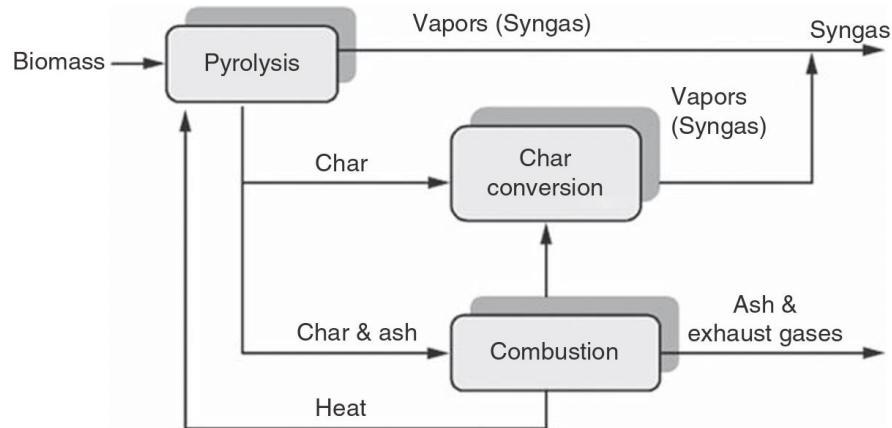


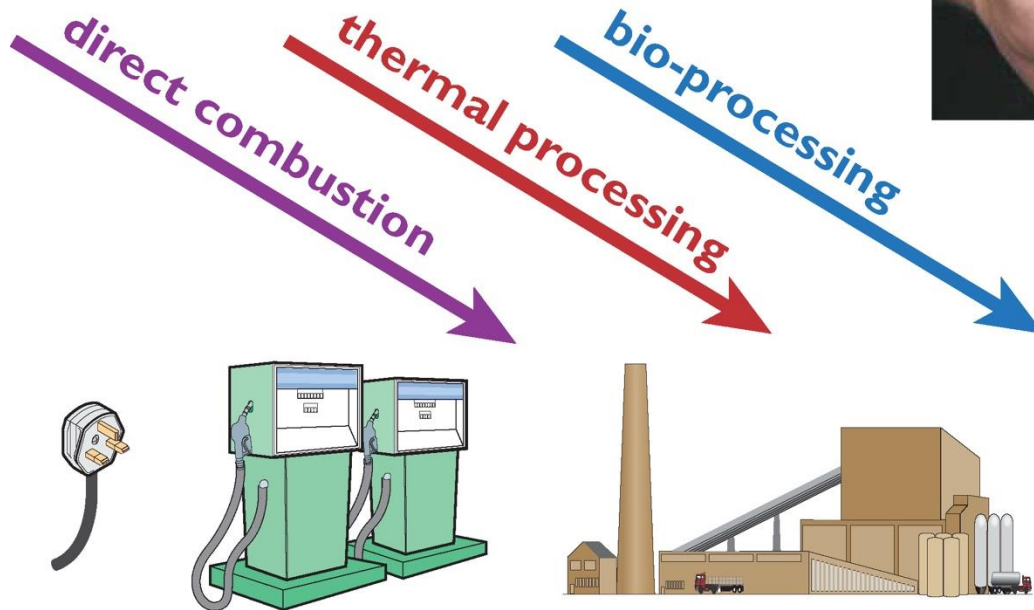
FIGURE 8.40 Biomass gasification process.

Syngas =
hydrogen (H_2) +
carbon monoxide (CO) +
methane (CH_4) +
carbon dioxide (CO_2) +
nitrogen (N_2) + tar

Biomass Conversion

To produce a fuel that is more conveniently used, transported or stored, or to exploit some property of the process itself.

- wood / forestry / arboricultural residues
- crops / straw / other agricultural residues
- perennial energy crops
- organic fraction of 'wastes' (including food / marine / aquatic)



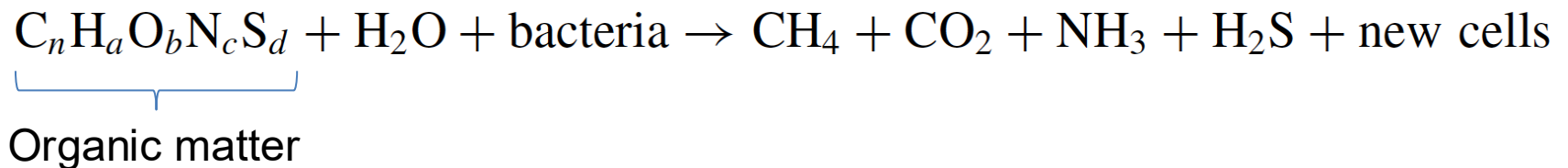
- Heat
- Electricity
- Hydrogen
- Other gases
- Ethanol / oils
- Chemicals
- Bio-based materials

Biomass Conversion (cont'd)

Thermal conversion: Use heat to convert biomass into another chemical form

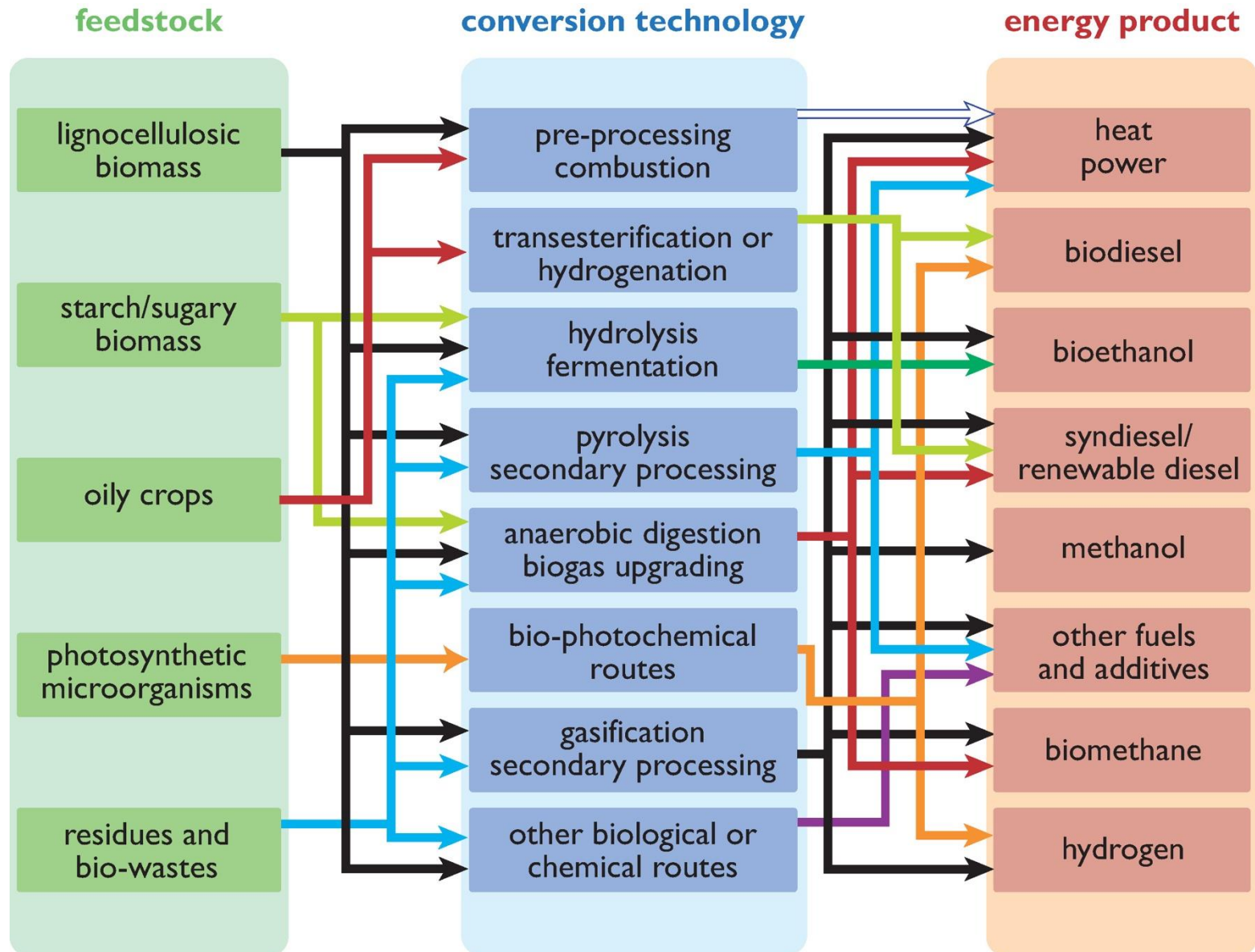
Chemical conversion: A range of chemical processes may be used to convert biomass into other forms

Biochemical conversion: Highly efficient biochemical processes have developed in nature to break down the molecules of which biomass is composed: *Anaerobic decomposition of organic materials by microorganisms to produce a biogas consisting primarily of methane and carbon dioxide.*

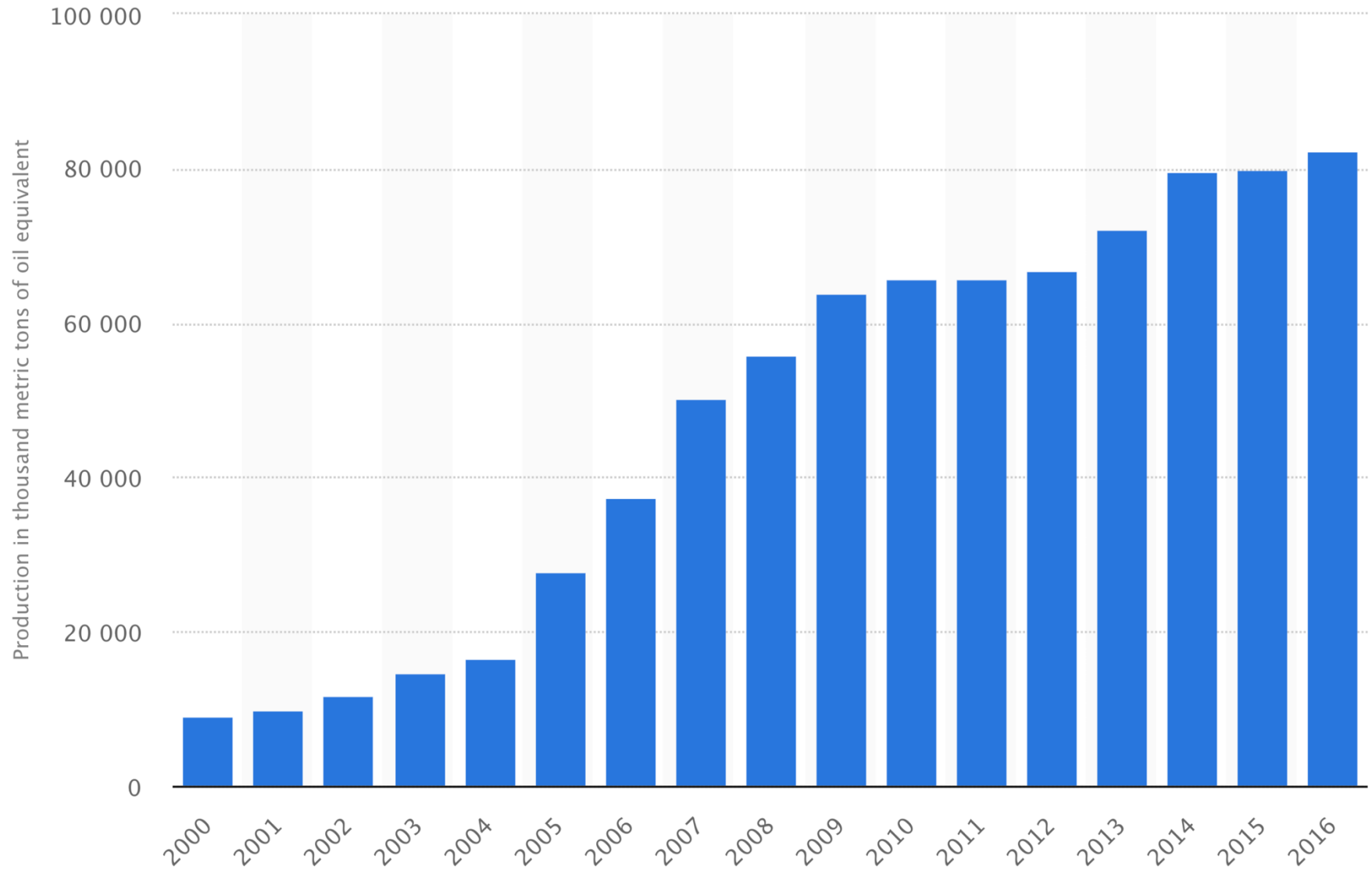


Electrochemical conversion: In addition to combustion, biomass/biofuels can be directly converted to electrical energy via electrochemical (electrocatalytic) oxidation of the material.

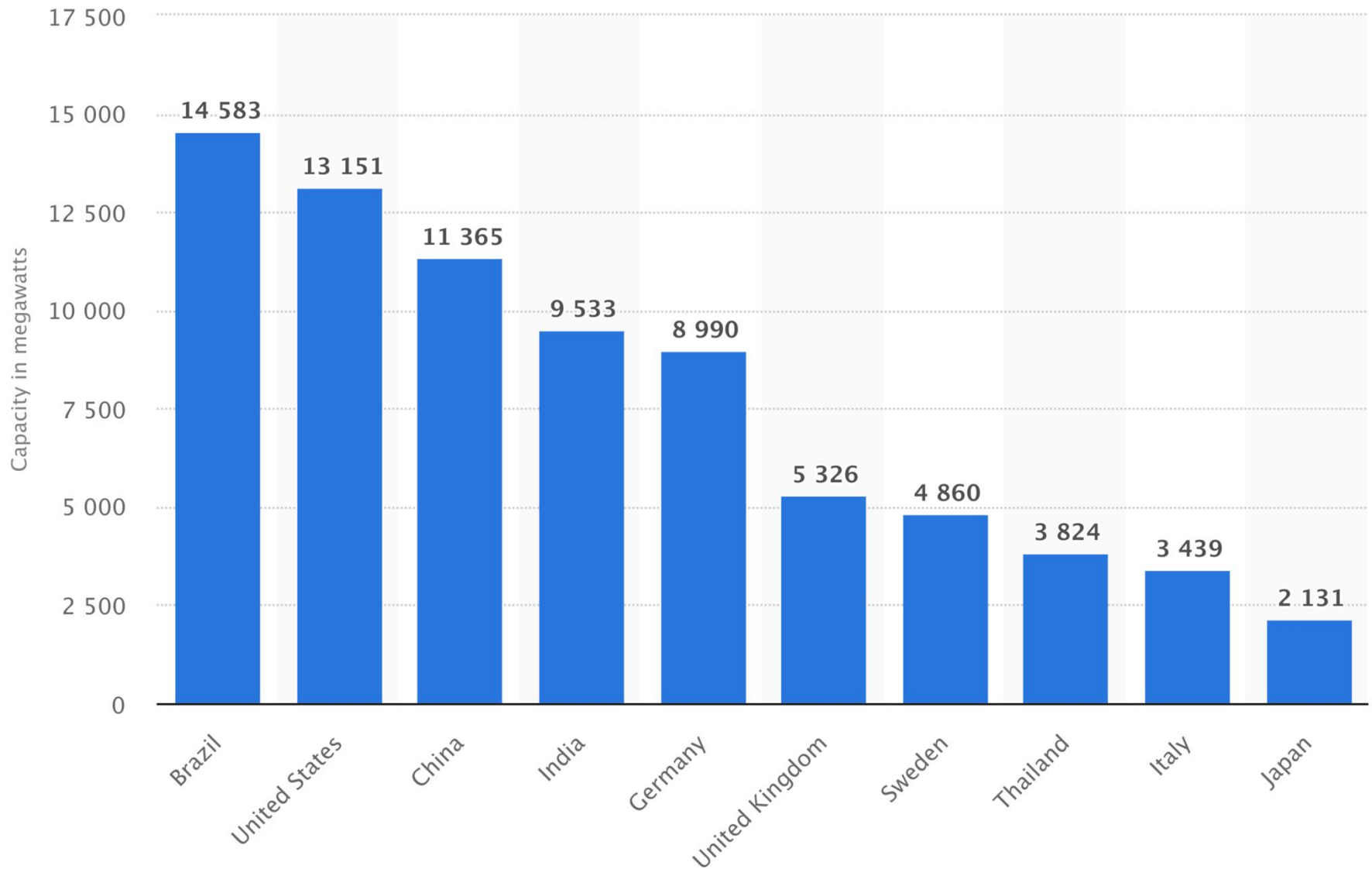
Biomass Conversion (Cont'd)



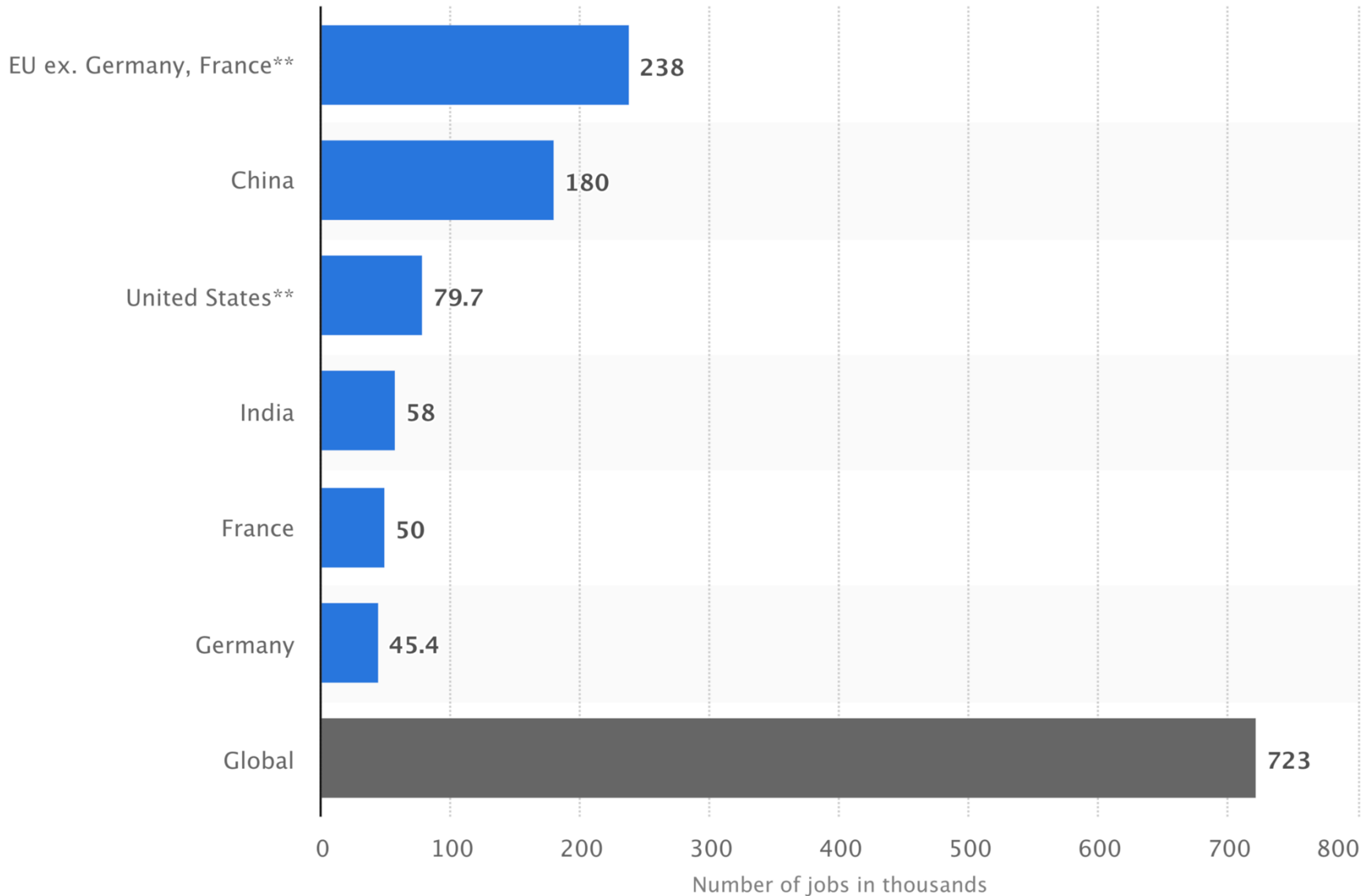
Global Biofuel Production



Leading Global Bioenergy Capacity in 2017



Number of Biomass Energy-related Jobs



Supply Chains

Supply chains play a key role in cost-effective delivery of bioenergy

Technical issues

- Inefficiencies of the conversion processes

Logistic issues

- Seasonal availability leading to storage challenges and/or seasonally idle facilities

Financial issues

- Lack of required transport infrastructure
- Limited flexibility or inflexibility to energy demand

Social issues

- Lack of participatory decision making
- Lack of public/community awareness

Policy and regulatory issues

- Lack of incentives to create competition among producers

Institutional/organizational issues

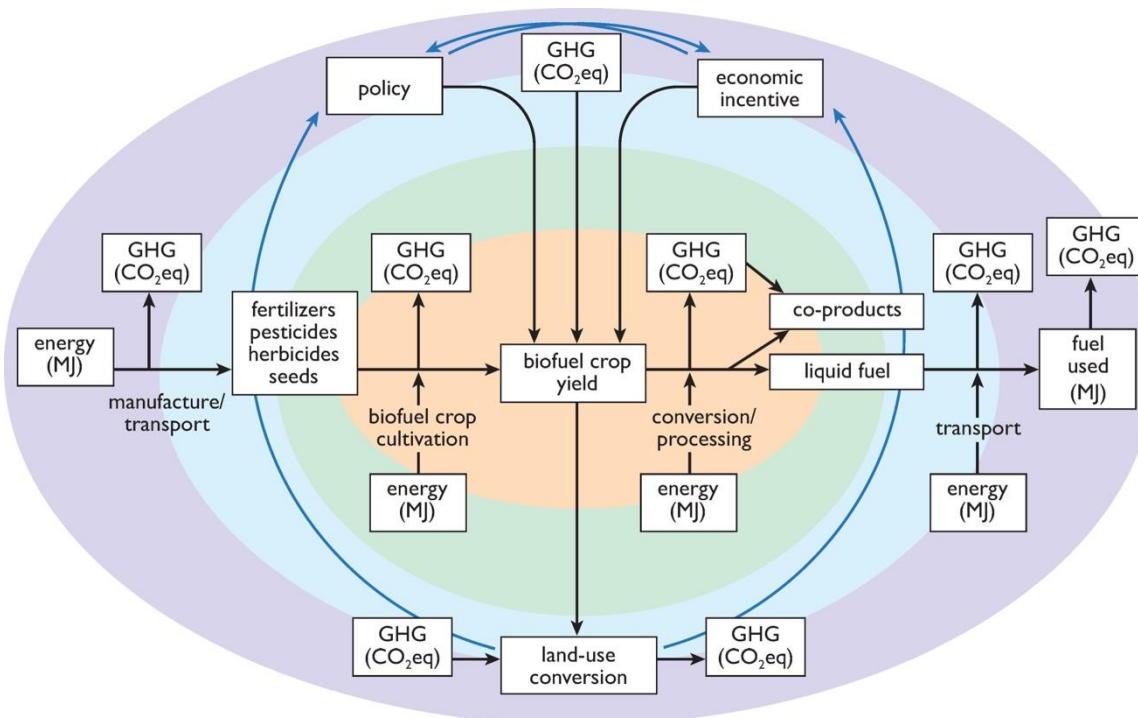
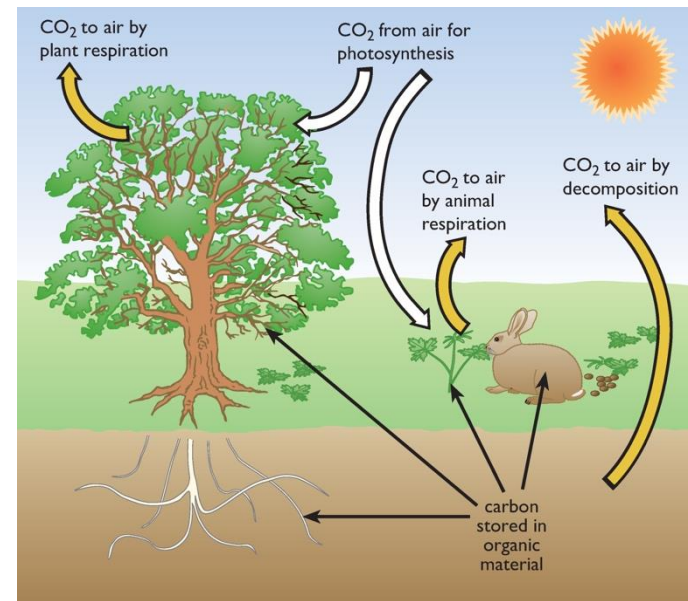
- Lack of supply chain standards

See details in <https://en.wikipedia.org/wiki/Biomass>

Environmental Impact

Using biomass produces air pollution in the form of

- Carbon monoxide and dioxide
- NO_x (nitrogen oxides)
- VOCs (volatile organic compounds)
- Particulates and other pollutants



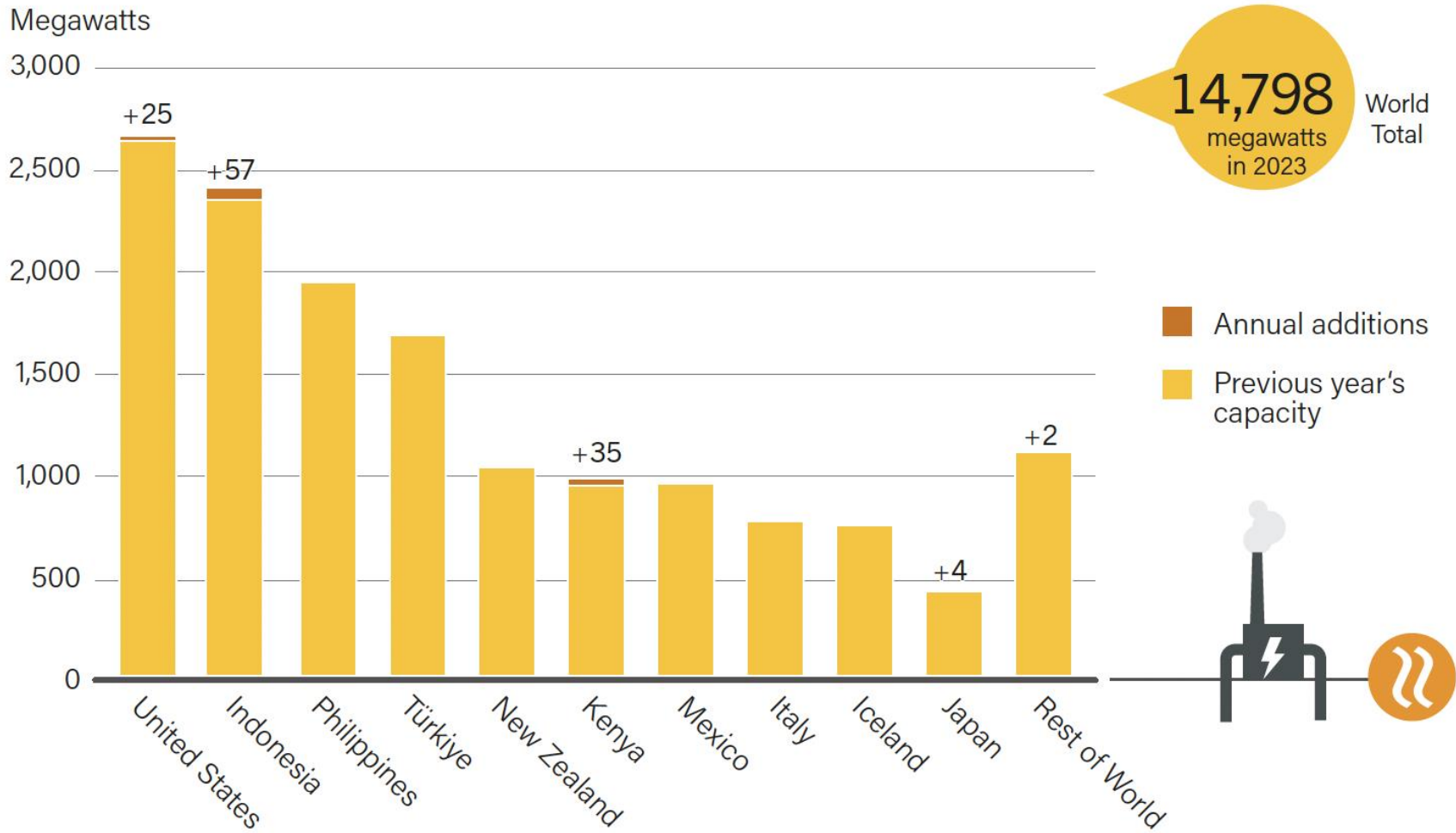
Different system boundaries for the life-cycle analysis of biomass energy systems

Geothermal Energy



Geothermal Development

Geothermal Power Capacity and Additions, Top 10 Countries and Rest of World, 2023

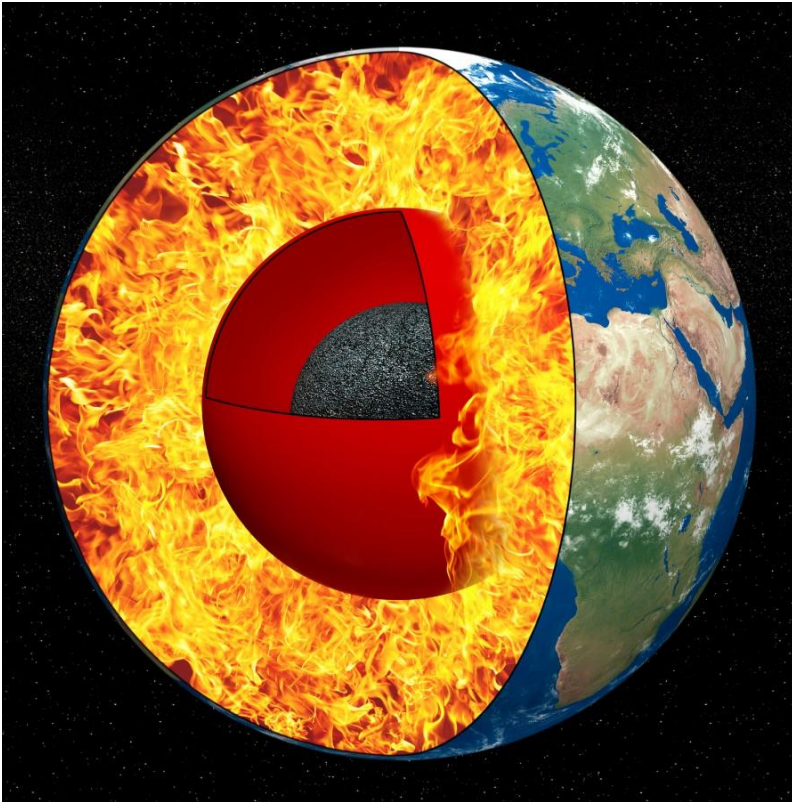


Hawaii Volcano Eruptions



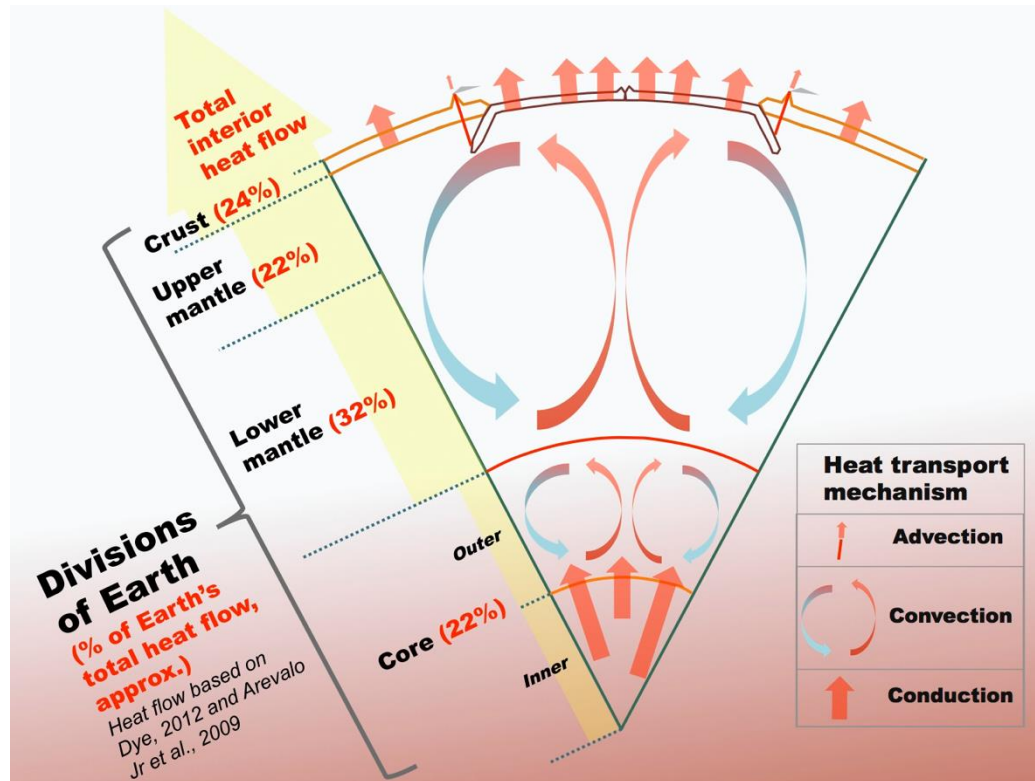
Geothermal Energy

Geothermal energy: Thermal energy generated and stored in the Earth



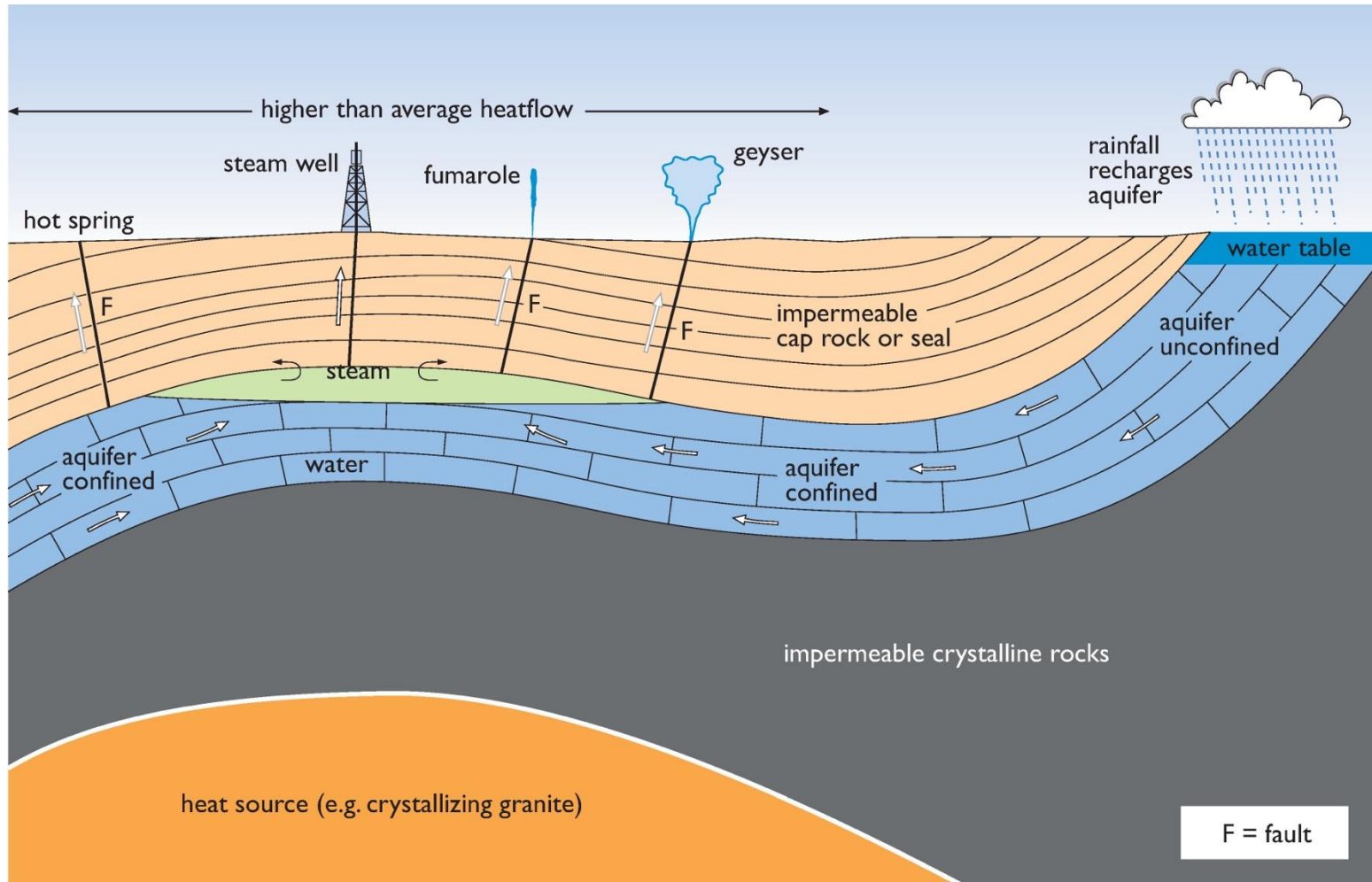
Earth Internal Heat

Earth's internal heat: Generated from radioactive decay and continual heat loss from Earth's formation



- Temperatures at the core–mantle boundary reach over 7200°F (4000°C)
- The high temperature and pressure cause rock to melt and solid mantle to behave plastically
- Rock and water are heated in the crust, up to 700°F (370°C)

The Physics of Geothermal Resources



- An aquifer containing water that can be accessed by drilling
- A cap rock to retain the geothermal fluid
- A heat source

Volcano-related Heat Sources

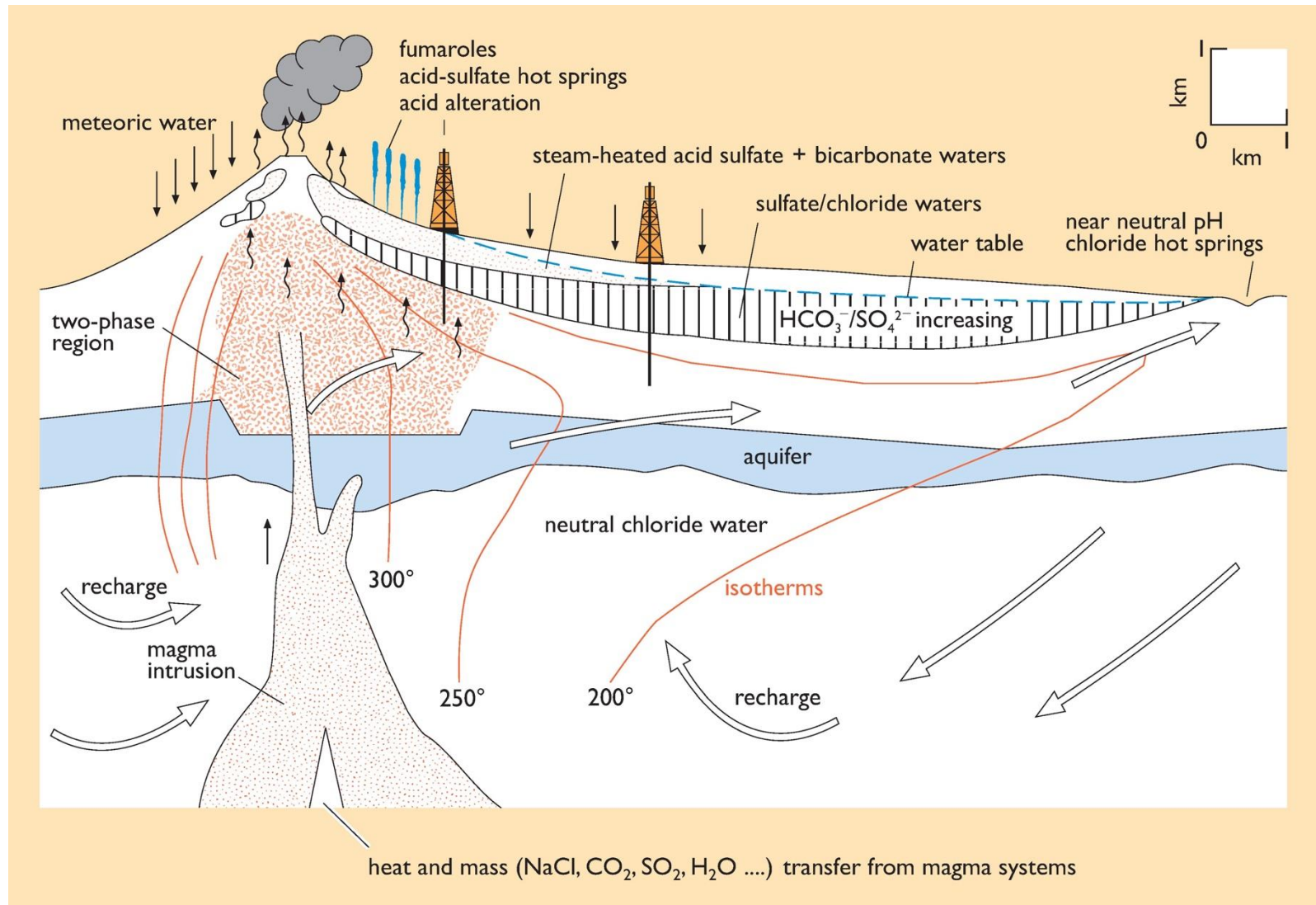
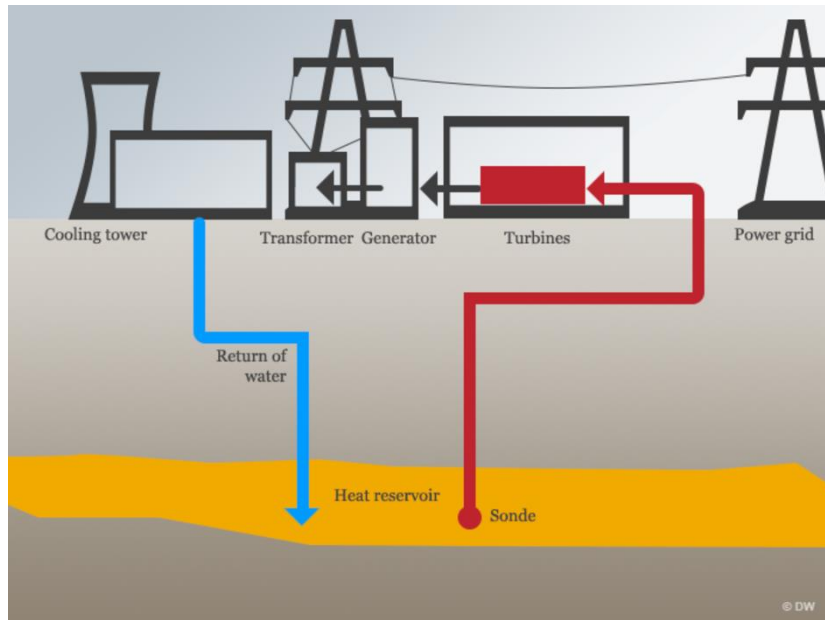


Fig: Conceptual model of a typical volcanic geothermal system

How Geothermal Energy Works



- Wells extract hot water or steam from geothermal reservoirs, driving turbines to produce electricity.
- Cooled water is then reinjected into the reservoir to sustain the cycle.
- This energy source offers consistent, low-impact electricity generation.

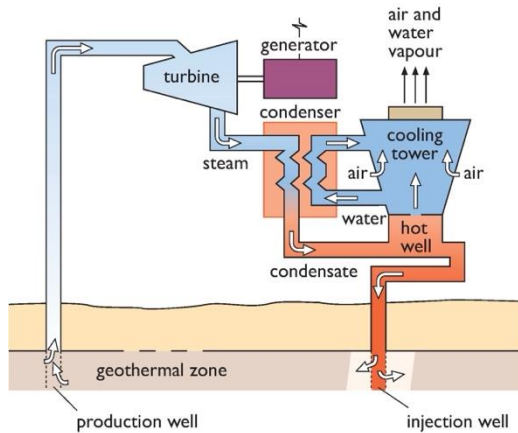
Production wells extract hot water and steam from geothermal reservoirs to the surface for electricity generation.

Injection wells return the cooled water back into the reservoir after it has passed through the power plant, maintaining pressure and ensuring the sustainability of the geothermal resource.

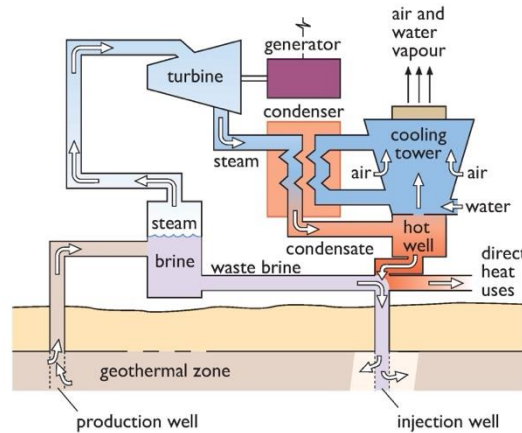
Production vs Injection Wells

Feature	Production Well	Injection Well
Primary Function	Extracts hot geothermal fluids	Returns cooled geothermal fluids to the reservoir
Location	Drilled into high-temperature geothermal reservoirs	Positioned away from production wells for effective reinjection
Operation	Brings heat to the surface for electricity generation	Reinforces reservoir pressure and sustainability
Temperature of Fluids	High-temperature fluids (above 150°C/300°F)	Cooled fluids (still warm but lower temperature)
Environmental Impact	Potential for resource depletion without proper management	Supports environmental sustainability and resource longevity
Role in Energy Cycle	Provides energy for electricity production	Completes the cycle by maintaining fluid levels and pressure

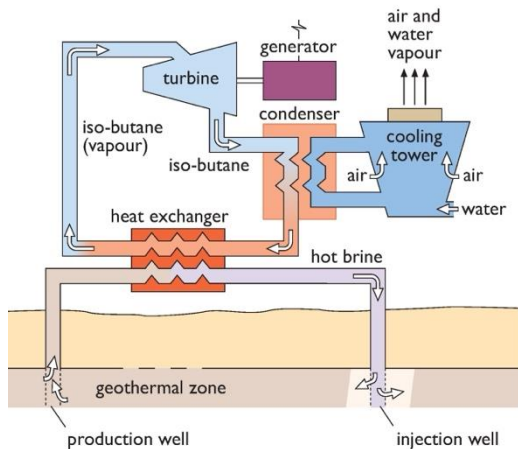
Dry Steam vs Flash Steam



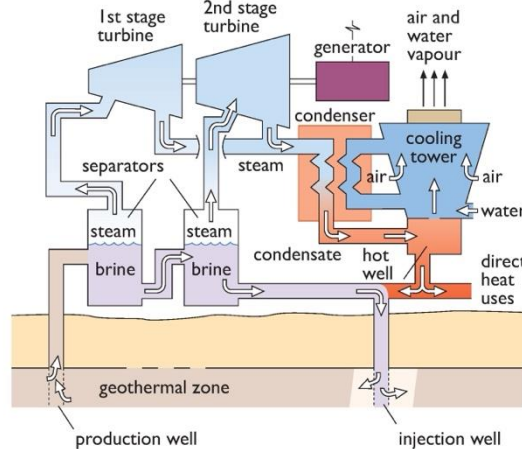
(a) dry steam power plant



(b) single flash steam power plant



(c) binary cycle / organic rankine cycle power plant

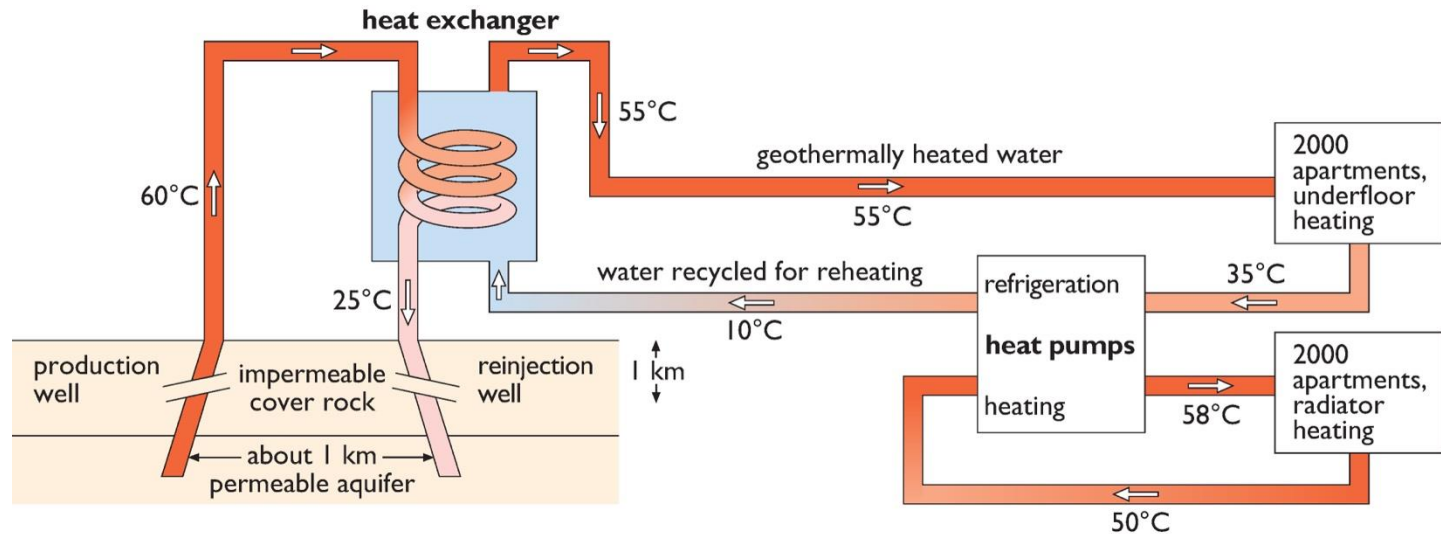


(d) double flash power plant

- Dry steam plants use natural steam directly from geothermal reservoirs to drive turbines, requiring high-temperature fields like The Geysers in CA.
- Flash steam plants extract hot water, reduce its pressure to create steam, and then use this steam to generate electricity, exemplified by Larderello in Italy. Flash steam plants are more complex due to the steam-water separation process but are also highly efficient.

Technologies for Direct Use of Geothermal

District heat systems

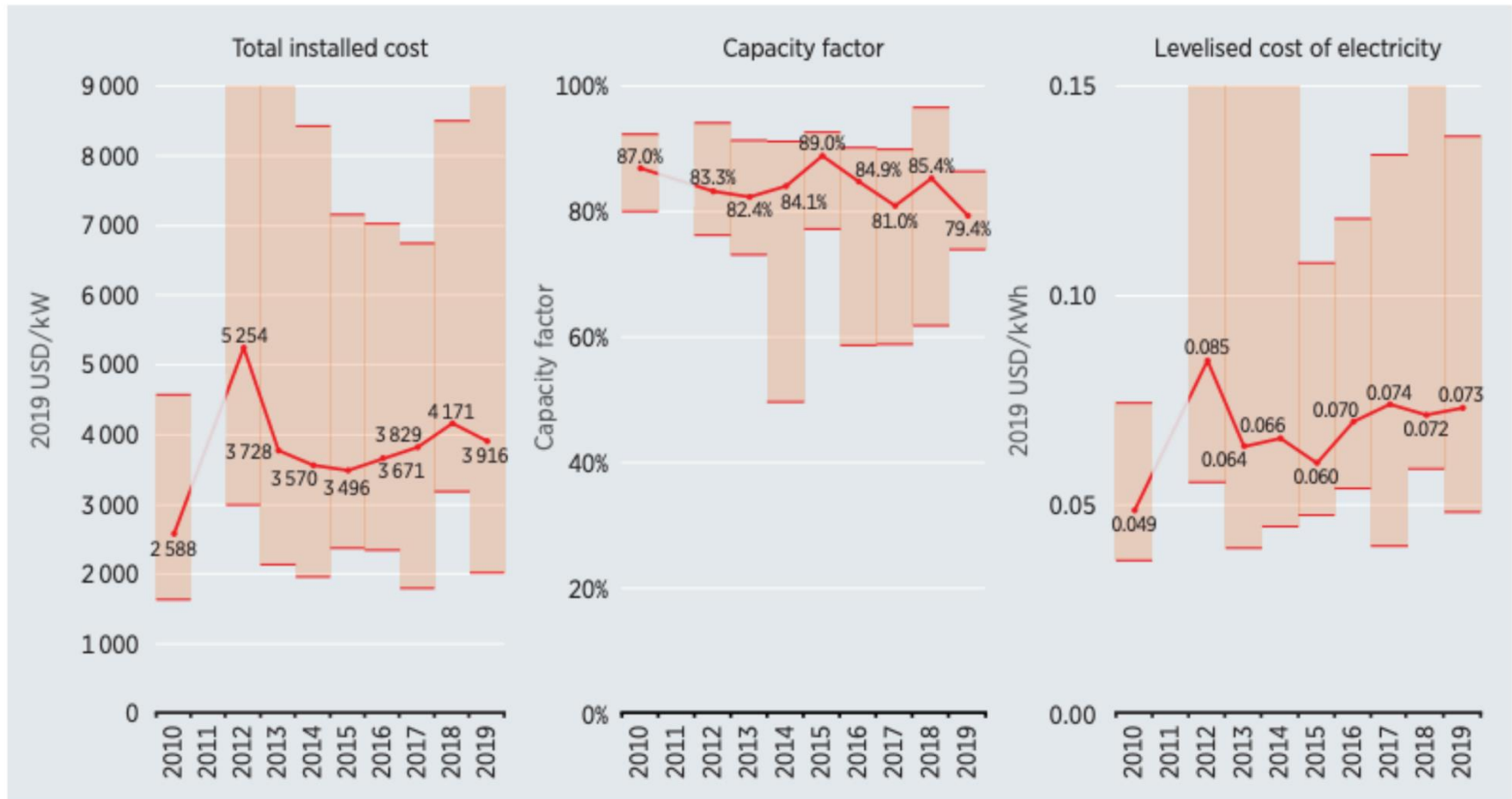


Enhanced geothermal systems



Costs of Geothermal

Global weighted average total installed costs, capacity factors and LCOE for geothermal, 2010-2019



Source: IRENA Renewable Cost Database.

LCOE of Geothermal Power

By technology and project size, 2007-2021

